Raman Spectroscopy and Machine Learning for Microplastics Identification and Classification in Water Environments

Yinlong Luo, Wei Su, Xiaobin Xu, Dewen Xu, Zhenfeng Wang, Hong Wu, Bingyan Chen, and Jian Wu

***Abstract*—As emerging pollutants of concern, microplastics (MPs) have been found in different water environments and have an impact on human health through the aquatic food chain. To advance our understanding of the traceability and environmental fate of MPs, reproducible and accurate methods, techniques, and analytical methods are necessary for MP type identification and characterization. In this study, based on Raman spectroscopy technology to extract characteristic peak information of MPs with fingerprint features, coupled to sparse autoencoder (SAE) and softmax classifier framework, the rapid identification and classification of six common MP (PET, PVC, PP, PS, PC, PE) particles in five water (pure water, rain water, lake water, tap water, and sea water) environments was realized. The results show that the average test accuracy of the trained algorithm is as high as 99.1%, which is better than 93.95% and 74.55% of the classical machine learning algorithms support vector machine (SVM) and back propagation (BP) neural network. Success rate indicates that the proposed method can be used to identify the MP samples.**

***Index Terms*—Microplastics, machine learning, raman spectroscopy, water environment.**

I. Introduction

IN RECENT years, microplastics (MPs) have attracted increasing attention as pollutants affecting the environment (water and soil) and human health [1]. MPs are the plastic fragments of ordinary products discharged into aquatic system, which are broken into small pieces of 5 mm to 1 μm due to erosion and decomposition under the influence of ultraviolet rays and microorganisms [2]. As plastic degradation proceeds, the total number of MPs increases exponentially. A large number of MPs, due to their small sizes, can enter animals and humans through the aquatic food chain, posing a potential threat to the global ecosystem, food safety and human health. Most of the MPs come from textiles, tires, water bottles, and even cosmetics [3]. For example, a large number of MPs are contained in hand sanitizers and facial scrubs used for exfoliation [4]. Therefore, the development of rapid and accurate identification and classification methods for different types of MPs plays an important role in the traceability of plastic pollutants in the water environment [5], [6].



Fig. 1. Common MPs and ways they are used.

Different plastic products often correspond to different chemical compositions, and most common MPs currently found in wastewater include polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), Polyvinyl chloride (PVC), polyethylene (PE), Polycarbonate (PC), and so on. Fig. 1 shows the various microplastics and their common applications. Different methods have been employed for analysis of MPs, including scanning electron microscopy, transmission electron microscopy, pyrolysis-gas chromatography/mass spectrometry, and non-destructive analytical methods as vibrational spectroscopy methods [7], [8], [9], [10]. Raman spectroscopy is a vibrational spectroscopy technology based on inelastic scattering of light [11], [12], [13]. Although Raman spectroscopy is difficult for quantitative measurements, individual substance has

Manuscript received 29 September 2022; revised 9 November 2022; accepted 10 November 2022. Date of publication 14 November 2022; date of current version 21 November 2022. This work was supported in part by the Fundamental Research Funds for Central Universities of China under Grant B210202150, in part by Nantong Science and Technology Program under Grant JC2021071, in part by the Natural Science Foundation of China under Grant 61705058, and in part by the Postgraduate Research & Practice Innovation Program of Jiangsu Province under Grant SJCX22_0164. *(Corresponding author: Wei Su.)*

Yinlong Luo, Wei Su, Xiaobin Xu, Dewen Xu, Zhenfeng Wang, and Bingyan Chen are with the College of Science, Hohai University, Changzhou 213022, China (e-mail: 211619010047@hhu.edu.cn; opticsu@hhu.edu.cn; xxbtc@hhu.edu.cn; 211619010084@hhu.edu.cn; wangzhenfeng@hhu.edu.cn; cbyan@foxmail.com).

Hong Wu is with the School of Science, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: wuhong@njupt.edu.cn).

Jian Wu is with the College of Advanced Interdisciplinary Studies, National University of Defense Technology, Changsha 410100, China (e-mail: wujian15203@163.com).

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/JSTQE.2022.3222065.

Digital Object Identifier 10.1109/JSTQE.2022.3222065

1077-260X © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

its own unique Raman characteristic peaks, which are similar to the fingerprint of chemical structure, so it is very suitable for quickly and accurately identifying and distinguishing similar samples [14], [15], [16].

However, due to the large amount of collected Raman spectral data and the complex chemical composition of most MPs, the extraction of qualitative or quantitative information is very difficult [17], [18], [19]. In this case, combining machine learning can help in the identification and classification of MPs [20], [21]. There are two main approaches known to be successful in identifying plastics. One is to obtain the appearance information of MPs by combining imaging methods such as hyperspectral imaging, holographic imaging and enhanced dark-field microscopy, and combined support vector machine (SVM), convolutional neural network and other algorithms to realize the classification of MP particles. However, due to the lack of analysis of information on the chemical composition of MPs, such methods cannot identify the type of MPs [22], [23], [24]. The other is to obtain the spectral information of plastics by combining spectroscopic techniques such as Fourier transform infrared spectroscopy, near-infrared reflectance spectroscopy, laser-induced breakdown spectroscopy, and X-ray fluorescence spectroscopy coupled with machine learning classifiers to achieve the identification of plastics. But at present, such methods are mostly used in the identification of marcoplastics [25], [26].

This study aims to use machine learning combined with Raman spectroscopy to achieve rapid analysis of MPs in different water environments. To our knowledge, this combination of technologies has been used for the first time in the identification and classification of MPs. This method is helpful to speculate on the origin of MPs and their possible human uses.

II. Materials and Methods

A. Chemicals and Materials

Six types of MP particles, PET, PP, PS, PVC, PE, and PC with a size of about 10 μm were purchased from Shanghai Fengtai Plastic & Chemical Co., Ltd. (Shanghai, China). Medium speed qualitative filter paper with pore size of about 15 μm to 20 μm was purchased from Xinxing Special Paper Co. Ltd (Hangzhou, China). $HAuCl_4 \cdot 3H_2O$ and $C_6H_5O_7Na_3$ were purchased from Sigma-Aldrich (Shanghai) Trading Co., Ltd (Shanghai, China), and all chemicals were of analytical grade and used without further purification. Ultrapure water system produced water with 18.2 MΩ/cm resistivity was used in all experiments. Since MPs have been detected in different water environments, and in order to verify the advantage of Raman spectroscopy to accurately identify target substances based on characteristic peaks even in mixed solutions containing impurities, five types of water (pure water, tap water, rain water, lake water and sea water) were used to disperse MPs separately, representing different water environments. Tap water was collected from the lab in the YingCai building in Hohai University. Rain water was collected during the heavy rain on July 26, 2022, in Changzhou, Jiangsu. Lake water was collected from the West Tai Lake (119°51′E, 31°41′N). Sea water was collected from Fuqing, Fujian (119°37′E, 25°21′N).

TABLE I
Functional Parameters of Confocal Raman Microscope System

Component	Model	Parameter	Value
Spectrometer	IDR-micro-785	Spectrum range	400~2200 cm^{-1}
Raman laser	SPL-laser-785	Wavelength	785 nm
		Power	20 mW
		Spectral linewidth	< 0.2 nm
		Spot size	~40 μm
Raman probe	SPL-rpb-785	optical density	46
Objective lens	SOPTOP-LMplan-50	Magnification	×50
		Numerical aperture	0.55

B. Raman Spectral Data Acquisition

MP particles were dispersed in different types of water and all obtained concentrations of 10 ppm, 1 ppm and 0.1 ppm. Since the concentration of MPs in the actual water environments are definitely different, thus three different concentrations were chosen to verify whether the proposed method is still accurate in the face of MP solutions with different concentrations. Considering that MP particles are difficult to disperse in water, surfactant (v/v = 1:200) was added to ensure uniform dispersion of MPs.

In a typical measurement, 2 mL MP solution was pipetted onto the filter paper based substrate for Raman measurements. In order to obtain stronger Raman characteristic peaks of MPs, a medium speed qualitative filter paper doped with gold nanoparticles was used as the detection substrate. Doping of gold nanoparticles can amplify the Raman signal through the enhancement of the local electric field. It is worth noting that the Raman measurements can be performed directly after the liquid was filtered, which can greatly reduce the pretreatment time. Details on substrate preparation and characterization are provided in Supplementary Materials.

A homemade confocal Raman microscope system was used to obtain MP Raman spectra. The specific functional parameters are shown in Table I.

C. Spectral Data Processing

Raman spectral data needs to be processed after acquisition, including baseline correction, smoothing and filtering, and spectral normalization. On the one hand, the preparation of the test samples and the parameter settings of the Raman spectral measurement system will affect the quality of the spectra; on the other hand, the processing of the obtained spectral data is of great help to the feature extraction in the subsequent algorithm.

1) Baseline Correction: For Raman measurements, the interference of the fluorescence background always exists and will cause the baseline drift of the obtained Raman spectra, which seriously affects the validity and accuracy of the spectral analysis. Therefore, the adaptive iteratively reweighted penalized least squares (airPLS) model was used for the baseline

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

correction [27], [28]. The airPLS is an approach which adopts an error-based iterative weighting strategy. The weight of each point is updated based on the difference between the baseline and the original signal of the last loop fitting, which is expressed as follows:

$$w_i^t = \begin{cases} 0 & x_i \geq z_i^{t-1} \\ \exp\left[\frac{t(x_i - z_i^{t-1})}{|d^t|}\right] & x_i < z_i^{t-1} \end{cases} \quad (1)$$

where x is the original signal, z is the fitted baseline, and d^t is the sum of the absolute values of less than variables in the t-th iteration x_i-z_i.

2) Smoothing and Filtering: There are many high-frequency noise signals in the initial Raman spectra, which can interfere with the algorithm to extract the main features of the spectra. Therefore, the Savitzky-Golay (SG) filter is used to smooth the noisy spectra. The SG filter performs least squares fitting on each window of data through a polynomial to obtain the smoothing coefficient h_i corresponding to each point in the window, which is expressed as follows [29]:

$$x_{k,\,\text{smooth}} = \bar{x}_k = \frac{1}{H} \sum_{i=-w}^{+w} x_{k+i} h_i \quad (2)$$

where $(2w+1)$ is the width of the filtering window, and $H = \sum_{i=-w}^{+w} h_i$ is the normalization factor.

3) Min-max Normalization: The purpose of min-max normalization is to linearly transform the original data to get the scaled data in the range (0, 1). The conversion formula is as follows [30]:

$$X^* = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (3)$$

where x, $x_{\max}$, $x_{\min}$ are the original spectral data, the maximum value in the spectral data, and the minimum value in the spectral data, respectively. The min-max normalization process is to perform center shift transformation and dimensionless compression on the Raman spectral data, which can effectively remove the influence of the laser light intensity, focusing effect and exposure time on the Raman signal intensity.

Taking PET as an example, Fig. 2 shows the original Raman spectra of PET and the Raman spectra after baseline correction, smoothing and filtering, and min-max normalization.

D. MPs Identification and Classification Using Machine Learning Method

Autoencoder (AE) is a type of feedforward neural network where the input is equal to the output. Fig. 3 shows the basic model, including an encoder and a decoder. The encoder maps the input to hidden representation, and the decoder tries to map the encoded features to reconstruct the original input. In general, the purpose of the model is to reduce high-dimensional raw data into low-dimensional data without losing information as much as possible. The feature of AE is that the number of nodes in the input layer (excluding bias nodes) is the same as the number of nodes in the output layer, while the number of nodes in the hidden layer is less than that in the input or output layer.

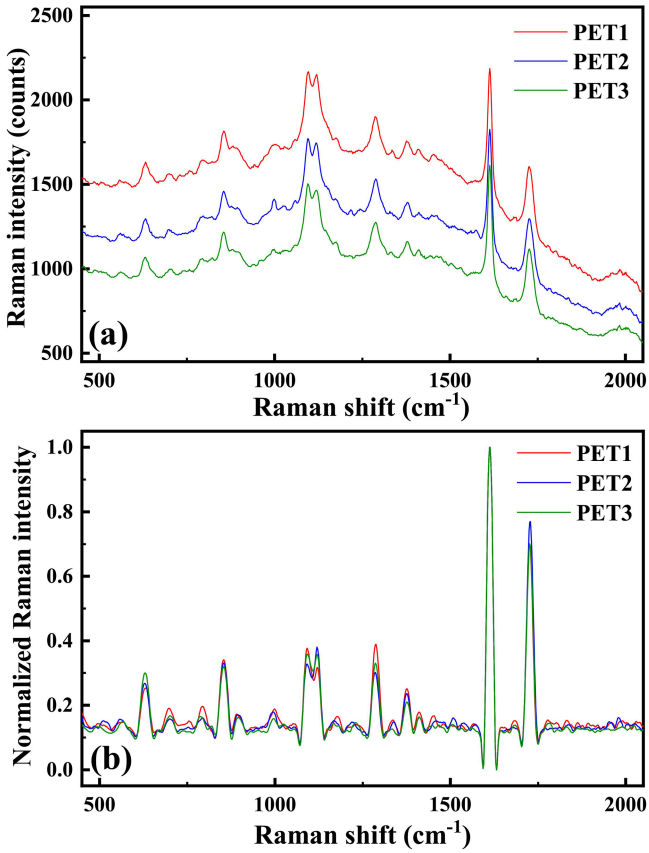


Fig. 2. (a) Original PET Raman spectra. (b) PET Raman spectra after baseline correction, smooth and filtering, and min-max normalization.

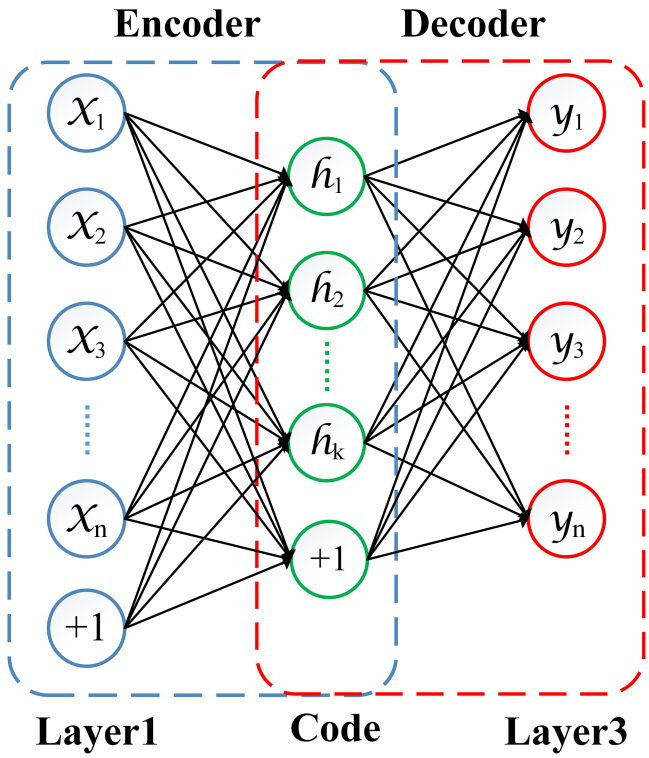


Fig. 3. Structure diagram of automatic encoder.

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

The Sparse-Autoencoder (SAE) is based on the AE with sparsity constraints, so that most of the nodes in the hidden layer are suppressed, and only a small fraction of nodes is activated [31]. Imposing sparsity constraints on the hidden layer enables it to learn features that can better express the samples in harsh environments, which is very suitable for feature extraction and identification of Raman spectra of MPs in different water environments. The average activation function of the hidden layers of SAE can be expressed as:

$$\hat{\rho}_j = \frac{1}{m}\sum_{i=1}^{m}\left[a_j^{(2)}(x^{(i)})\right] \tag{4}$$

where $a_j(x^{(i)})$ is the activation level of the *j*-th neuron in the hidden layer. Due to the sparsity requirement, it is expected that most hidden neurons will not be activated, i.e., the average activation tends to be a constant close to zero. Hence, the L1 regularization term as an additional penalty term is added to the loss function, and use the hyperparameter β to control its proportion in the loss function. The resulting loss function is as follows:

$$J_{\text{sparse}}(w,b) = J(w,b) + \beta\sum_{j=1}^{s_2} KL(\rho||\hat{\rho}_j) \tag{5}$$

where *J*(*w*, *b*) is the loss function without sparsity constraints. $KL(\rho||\hat{\rho}_j)$, as an additional sparsity penalty, is the relative entropy between two Bernoulli random variables with mean ρ and mean $\hat{\rho}_j$. The mathematical expression of $KL(\rho||\hat{\rho}_j)$ is as follows:

$$KL(\rho||\hat{\rho}_j) = \rho\ln\frac{\rho}{\hat{\rho}_j} + (1-\rho)\ln\frac{1-\rho}{1+\hat{\rho}_j} \tag{6}$$

III. Results and Discussions

Fig. 4 shows the basic training framework of SAE for identifying MP species. Firstly, the collected Raman spectral data are preprocessed to establish the Raman spectra database of MPs. Then, 20% of the spectral data is randomly selected from the database as the testing set, and the remaining data is used as the training set. After the algorithm model for MP identification is trained, the testing set is used to test the effect of the trained model on the MP types identification.

A. Raman Spectra Database of MPs

In order to be able to accurately predict the types of MPs, a standard Raman spectra database of MPs needs to be established. First, dispersed MP particles (PET, PVC, PP, PS, PC, PE) in water sources (pure water, rain water, lake water, tap water, and sea water), and obtained three concentrations of 10 ppm, 1 ppm and 0.1 ppm, respectively. Then dropped each analyte solution on the prepared filter paper based detection substrate for Raman spectroscopy measurement. To ensure the reliability of the experimental results, each kind of analyte solution was dropped on 7 Raman detection substrates prepared by the same method. In addition, the database also includes the Raman spectra of the blank controls, that is, five different water sources were directly added to the substrate for Raman measurement. For the above

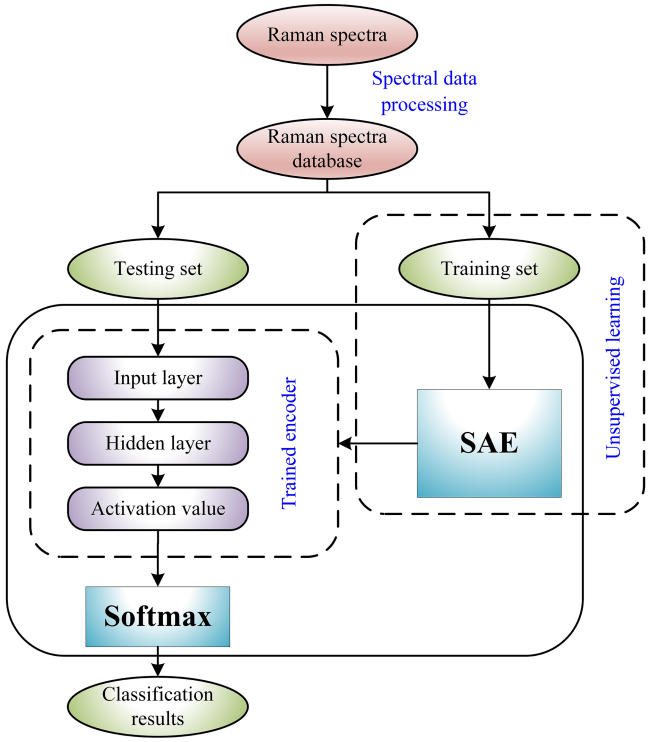


Fig. 4. Basic frame structure of SAE model.

735 samples, the Raman spectra of 5 random points on each sample were measured, and finally 3675 Raman spectra were collected. After performing baseline correction, smooth and filtering, and min-max normalization for each Raman spectrum, the establishment of the Raman spectra database was completed.

The identification of MPs by combining machine learning and Raman spectroscopy is based on the different chemical structures of different MPs, so the chemical bonds contained are not exactly the same, which means the positions of the Raman characteristic peaks are not exactly the same. Since most of the characteristic peaks of MPs are located from 500 cm^{-1} to 1800 cm^{-1}, the Raman spectral data in this wavenumber range is mainly intercepted when creating the database. Fig. 5 shows the Raman spectra of PET, PVC, PP, PS, PC and PE, and their main characteristic peaks. Main Raman characteristic peaks of different types of MPs and their assignments are shown in Table II [32], [33], [34], [35], [36], [37], [38], [39]. The specific Raman spectra of different MPs in different water environments in database are shown in Figs. S2-S7.

B. Machine Learning for the MPs' Investigation

In this article, a model including SAE is built on PyCharm software platform. Table III shows the main hyperparameters and structure of SAE. The SAE model includes a hidden layers with a size of 30 nodes, and the maximum training times is 3000. The activation functions of the model are ReLU and Softmax functions, where the activation function of the classifier is Softmax, and the activation functions of other layers is ReLU. In regularization, considering L2 regularization, the solution

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

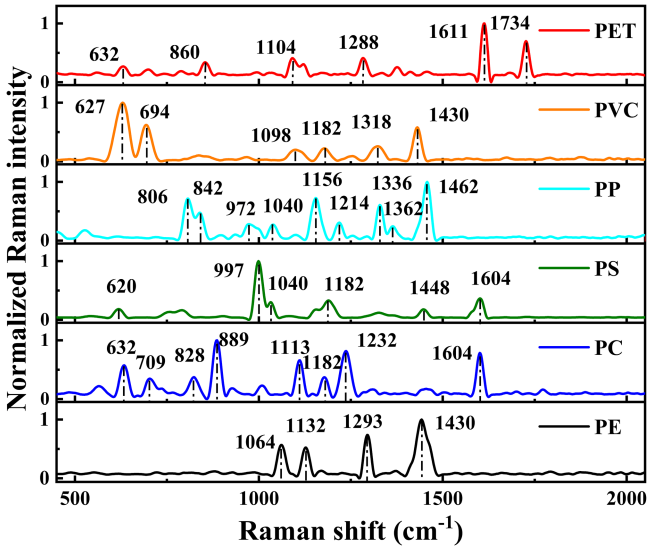


Fig. 5. Raman spectra of 6 types of MPs and their Raman characteristic peaks.

TABLE II
MAIN RAMAN CHARACTERISTIC PEAKS OF DIFFERENT TYPES OF MPS AND THEIR ASSIGNMENTS

Types of MPs	Wavenumber (cm^{-1})	Assignments
PET	632	Symmetric bending C=C
	860, 1288, 1611	Stretching C=C
	1104, 1288	Stretching C–O
	1734	Stretching C=O
PVC	627, 694	Stretching C–Cl
	1430	Stretching C–C
	1098, 1318	Deformation
PP	806, 972, 1040, 1156, 1214,	Stretching C–C
	842	Rocking CH_2
	1336	Bending CH
	1362	Symmetric bending CH_3
	1462	Bending CH_2
PS	620	Deformation
	997	Ring breathing
	1040	In-plane deformation C–H
	1182	Stretching C–C
	1448	Scissoring CH_2
	1604	Stretching ring-skeletal
PC	632	Deformation
	709	Out-of-plane bending CH
	828	Stretching ring
	889	Stretching C–CH_3
	1113	In-plane bending CH
	1182	Stretching C–O–C
	1232	Asymmetric stretching C–O–C
	1604	Stretching ring
PE	1064, 1132	Stretching C–C
	1293	Twisting CH_2
	1430	Bending CH_2

TABLE III
MAIN HYPERPARAMETERS AND STRUCTURE OF SAE

Structure/Hyperparameter	Numerical value/type
Number of Hidden Layers	1
Number of Hidden Nodes	30
Activation Function	ReLU Softmax
Penalty Weight β	5
Sparse Ratio	0.1
L2 Weight	0.006

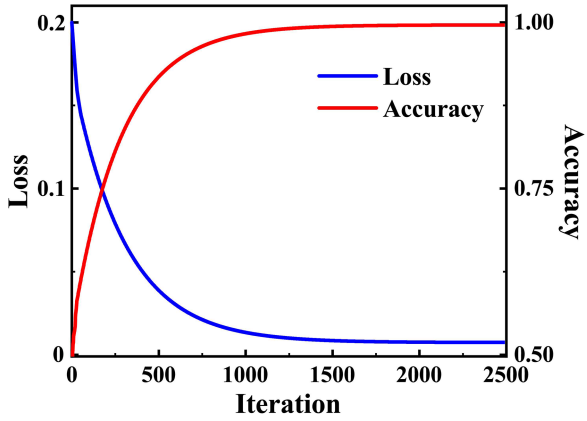


Fig. 6. Training loss and accuracy obtained with the dropout layer deactivated after training at each iteration.

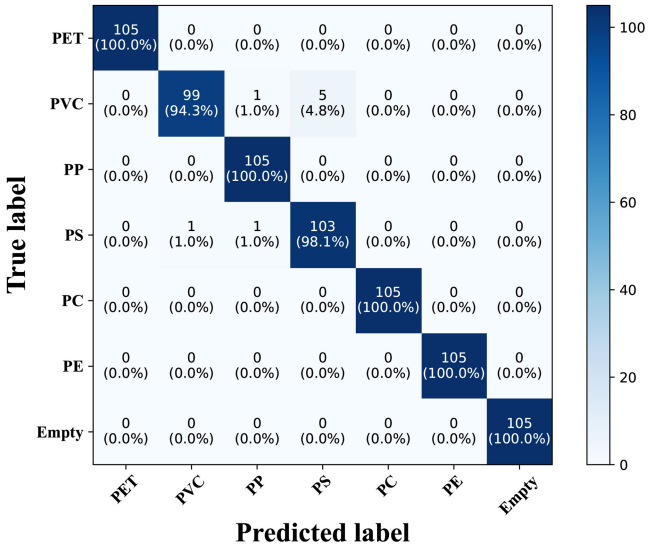


Fig. 7. Confusion matrix for cross-validation results from SAE model.

obtained is relatively smooth. Meanwhile, it can ensure that there are many dimensions close to 0 in the solution and reduce the complexity of the model. L2 regularization is selected, and its weight on the network is set to 0.006. However, L2 regularization cannot obtain the sparse model, so it is necessary to introduce a sparse penalty term, and its influence coefficient is set to 5. Finally, sparse ratio means that the expected proportion of neurons responding to the training sample is set to 0.1.

After setting the parameters and structure, the next step is to input the Raman spectra of the training set to train the model. Fig. 6 shows the variation of the loss function and classification accuracy with the number of iterations. It can be seen that after 1500 iterations, the loss function converges and the classification accuracy is close to 100%.

The created database has a total of 3675 Raman spectra, 20% samples were chosen as the testing set. Fig. 7 shows the confusion matrix results based on the SAE model. The number and percentage of the total number of objects is displayed in each cell. In the confusion matrix, rows and columns correspond to

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

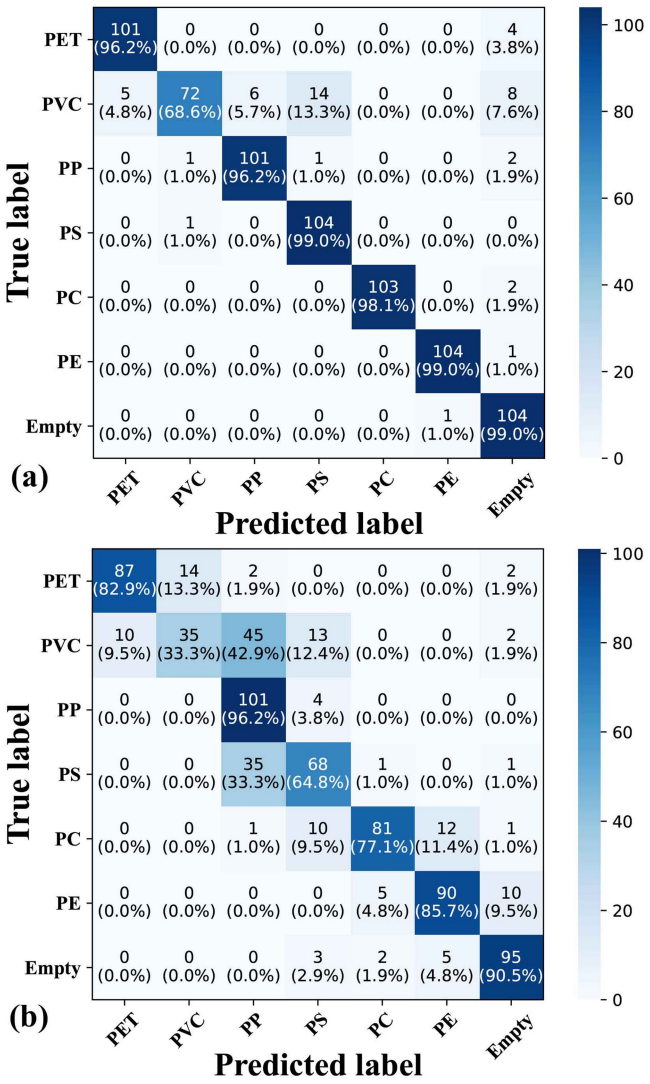


Fig. 8. (a) Confusion matrix for cross-validation results from SVM model. (b) Confusion matrix for cross-validation results from BP neural network.

actual categories and prediction categories, respectively. Diagonal cells correspond to correctly classified observations. Non diagonal cells correspond to observations of misclassification. As can be seen from Fig. 7, only one sample from PVC was wrong attributed to PP, five samples from PVC were wrong attributed to PS, one sample from PS was wrong attributed to PP, and one sample from PS was wrong attributed to PVC. The test accuracy is as high as 99.1%.

C. Comparison With Other Machine Learning Methods

To verify the superiority of SAE in the classification of MPs, SVM algorithm and back propagation (BP) neural network are used to train and recognize the Raman spectra of MPs. SVM algorithm and BP neural network, as classic machine learning algorithms, combined with Raman spectroscopy technique have been proved to have good accuracy in detecting pathogenic bacteria, cancer cells, rice, etc [40], [41], [42], [43], [44].

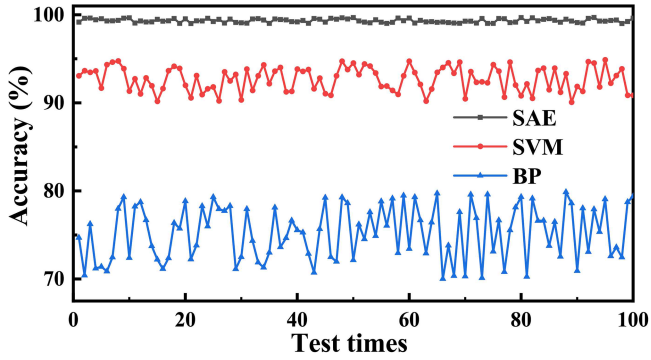


Fig. 9. Comparison of classification results of MPs by three methods after 100 tests.

Fig. 8 shows the confusion matrices of SVM and BP neural network. In SVM and BP neural network training classification, the division of sample training set and testing set is the same as SAE. The overall accuracy of SVM recognition is 93.7%, which is slightly lower than SAE, while the overall accuracy of BP neural network is only 75.8%, much lower than SVM and SAE.

In order to verify the stability of the model and further compare three classification methods, three models were tested 100 times respectively. Fig. 9 shows the accuracy comparison of the three methods after 100 tests. For our proposed method, the maximum accuracy can reach 100%, and the average accuracy is 99.51%, which shows great accuracy and excellent stability. The highest accuracy of SVM is 96.62%, and the average accuracy is only 93.95%, while the highest accuracy of BP neural network is only 81.10%, and the average accuracy is 74.55%. It can be seen that the unsupervised machine learning combined with SAE and Softmax classifier has superior performance in terms of stability and accuracy comparing with SVM and BP neural network.

The method proposed in this article is mainly aimed at the identification and classification of different MPs. If we want to identify the concentration of MPs in the water environments, in theory, we only need to omit the normalization step during the spectral data processing. However, due to the limitations of Raman spectroscopy technique in quantitative measurements, it may have higher requirements for measurement consistency, such as ensuring the consistency of light source and spectrometer parameters.

IV. Conclusion

In this study, a new approach for MPs identification and classification was developed and successfully tested, being based on the rapid processing of Raman spectra using machine learning algorithms. Distinct varieties of MPs (PET, PP, PS, PVC, PE, and PC) dispersed in different water environments (pure water, tap water, rain water, lake water and sea water) with concentrations of 10 ppm, 1 ppm and 0.1 ppm were included in this study. Baseline correction, smoothing and filtering, and min-max normalization were used to process the collected spectra, which can effectively remove the interference of impurities in the water source and the influence of the MP concentration. With SAE

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

and Softmax classifier framework, MP types were identified with 99.1% success rates, which is better than SVM and BP neural network. The whole process can be divided into MP Raman spectral database acquisition, spectral data processing, model training, identification and classification, the time spent on Raman measurements and data processing is about 5 hours. And after about 15 minutes of model training, the time spent in the MP identification and classification testing process is almost negligible, which shows the characteristics of rapid analysis. The proposed method for MPs identification and classification can play an important role in the traceability of plastic pollutants in the water environments.

REFERENCES

[1] M. Piccardo et al., "PET microplastics toxicity on marine key species is influenced by pH, particle size and food variations," *Sci. Total Environ.*, vol. 715, May 2020, Art. no. 136947.

[2] H. W. Cai et al., "Analysis of environmental nanoplastics: Progress and challenges," *Chem. Eng. J.*, vol. 410, 2021, Art. no. 128208.

[3] C. Schmid, L. Cozzarini, and E. Zambello, "Microplastic's story," *Mar. Pollut. Bull.*, vol. 162, Jan. 2021, Art. no. 111820.

[4] J. Gago, F. Galgani, T. Maes, and R. C. Thompson, "Microplastics in seawater: Recommendations from the marine strategy framework directive implementation process," *Front. Mar. Sci.*, vol. 3, Nov. 2016, Art. no. 219.

[5] A. Turner and L. Holmes, "Occurrence, distribution and characteristics of beached plastic production pellets on the island of Malta (central Mediterranean)," *Mar. Pollut. Bull.*, vol. 62, no. 2, pp. 377–381, Feb. 2011.

[6] Y. K. Song, S. H. Hong, M. Jang, G. M. Han, and W. J. Shim, "Occurrence and distribution of microplastics in the Sea surface microlayer in jinhae bay, South Korea," *Arch. Environ. Con, Tox.*, vol. 69, no. 3, pp. 279–287, Oct. 2015.

[7] A. Käppler et al., "Analysis of environmental microplastics by vibrational microspectroscopy: FTIR, Raman or both?," *Anal. Bioanal. Chem.*, vol. 408, no. 29, pp. 8377–8391, Nov. 2016.

[8] S. M. Patil et al., "An assessment of micro- and nanoplastics in the biosphere: A review of detection, monitoring, and remediation technology," *Chem. Eng. J.*, vol. 430, 2022, Art. no. 132913.

[9] S. Veerasingam et al., "Contributions of Fourier transform infrared spectroscopy in microplastic pollution research: A review," *Crit. Rev. Environ. Sci. Tec.*, vol. 51, no. 22, pp. 2681–2743, 2021.

[10] G. J. Xu et al., "Surface-enhanced Raman spectroscopy facilitates the detection of microplastics <1 μm in the environment," *Environ. Sci. Tech.*, vol. 54, no. 24, pp. 15594–15603, Dec. 2020.

[11] J. Levermore, T. Smith, F. Kelly, and S. Wright, "Detection of microplastics in ambient particulate matter using Raman spectral imaging and chemometric analysis," *Anal. Chem.*, vol. 92, no. 13, pp. 8732–8740, 2020.

[12] Y. Jeon et al., "Detection of nanoplastics based on surface-enhanced Raman scattering with silver nanowire arrays on regenerated cellulose films," *Carbohyd. Polym.*, vol. 272, Nov. 2021, Art. no. 118470.

[13] L. Lv et al., "In situ surface-enhanced Raman spectroscopy for detecting microplastics and nanoplastics in aquatic environments," *Sci. Total Environ.*, vol. 728, Aug. 2020, Art. no. 138449.

[14] J. T. Lu, Q. S. Xue, H. X. Bai, and N. Wang, "Design of a confocal micro-Raman spectroscopy system and research on microplastics detection," *Appl. Opt.*, vol. 60, no. 27, pp. 8375–8383, Sep. 2021.

[15] J. Caldwell, P. Taladriz-Blanco, B. Rothen-Rutishauser, and A. Petri-Fink, "Detection of Sub-Micro- and Nanoplastic particles on gold nanoparticle-based substrates through surface-enhanced Raman scattering (SERS) spectroscopy," *Nanomater.-Basel*, vol. 11, no. 5, Apr. 2021, Art. no. 1149.

[16] C. F. Araujo, M. M. Nolasco, A. M. P. Ribeiro, and P. J. A. Ribeiro-Claro, "Identification of microplastics using Raman spectroscopy: Latest developments and future prospects," *Water Res.,* vol. 142, pp. 426–440, Oct. 2018.

[17] C. Berghian-Grosan and D. A. Magdas, "Application of Raman spectroscopy and machine learning algorithms for fruit distillates discrimination," *Sci. Rep.*, vol. 10, no. 1, Dec. 2020, Art. no. 21152.

[18] J. H. Cao et al., "Extraction and identification methods of microplastics and nanoplastics in agricultural soil: A review," *J. Environ. Manage.*, vol. 294, 2021, Art. no. 112997.

[19] E. Ryzhikova et al., "Raman spectroscopy and machine learning for biomedical applications: Alzheimer's disease diagnosis based on the analysis of cerebrospinal fluid," *Spectrochim. Acta. A*, vol. 248, Mar. 2021, Art. no. 119188.

[20] G. Renner et al., "Data preprocessing & evaluation used in the microplastics identification process: A critical review & practical guide," *TrAC Trend. Anal. Chem.*, vol. 111, pp. 229–238, Feb. 2019.

[21] H. D. M. Back, E. C. Vargas Junior, O. E. Alarcon, and D. Pottmaier, "Training and evaluating machine learning algorithms for ocean microplastics classification through vibrational spectroscopy," *Chemosphere*, vol. 287, Jan. 2022, Art. no. 131903.

[22] J. Shan et al., "A novel way to rapidly monitor microplastics in soil by hyperspectral imaging technology and chemometrics," *Environ. Pollut.*, vol. 238, pp. 121–129, Jul. 2018.

[23] Y. Zhu, C. Yeung, and E. Lam, "Microplastic pollution monitoring with holographic classification and deep learning," *J Phys-Photon.*, vol. 3, no. 2, 2021, Art. no. 024013.

[24] I. Ishmukhametov, L. Nigamatzyanova, G. Fakhrullina, and R. Fakhrullin, "Label-free identification of microplastics in human cells: Dark-field microscopy and deep learning study," *Anal. Bioanal. Chem.*, vol. 414, no. 3, pp. 1297–1312, Jan. 2022.

[25] D. Stefas, N. Gyftokostas, E. Bellou, and S. Couris, "Laser-Induced breakdown spectroscopy assisted by machine learning for plastics/polymers identification," *Atoms*, vol. 7, no. 3, Aug. 2019, Art. no. 79.

[26] A. P. M. Michel et al., "Rapid identification of marine plastic debris via spectroscopic techniques and machine learning classifiers," *Environ. Sci. Tech.*, vol. 54, no. 17, pp. 10630–10637, Sep. 2020.

[27] Z. M. Zhang, S. Chen, and Y. Z. Liang, "Baseline correction using adaptive iteratively reweighted penalized least squares," *Analyst*, vol. 135, no. 5, pp. 1138–1146, May 2010.

[28] Z. M. Zhang et al., "An intelligent background-correction algorithm for highly fluorescent samples in Raman spectroscopy," *J. Raman Spectrosc.*, vol. 41, no. 6, pp. 659–669, Oct. 2010.

[29] A. Savitzky and M. Golay, "Smoothing and differentiation of data by simplified least squares procedures," *Anal. Chem.*, vol. 36, no. 8, pp. 1627–1639, 1964.

[30] V. Gajera, R. G. Shubham, and P. K. Jana, "An effective multi-objective task scheduling algorithm using min-max normalization in cloud computing," in *Proc. IEEE 2nd Int. Conf. Appl. Theor. Comput. Commun. Technol.*, 2016, pp. 812–816.

[31] M. Al-Qatf, Y. Lasheng, M. Al-Habib, and K. Al-Sabahi, "Deep learning approach combining sparse autoencoder with SVM for network intrusion detection," *IEEE Access*, vol. 6, pp. 52843–52856, 2018.

[32] M. Mazilu, A. C. D. Luca, A. Riches, C. S. Herrington, and K. Dholakia, "Optimal algorithm for fluorescence suppression of modulated Raman spectroscopy," *Opt. Exp.*, vol. 18, no. 11, pp. 11382–11395, May 2010.

[33] A. S. Nielsen, D. N. Batchelder, and R. Pyrz, "Estimation of crystallinity of isotactic polypropylene using Raman spectroscopy," *Polymer*, vol. 43, no. 9, pp. 2671–2676, 2002.

[34] C. Zimmerer et al., "Nondestructive characterization of the polycarbonate - octadecylamine interface by surface enhanced Raman spectroscopy," *Polym. Test.*, vol. 73, pp. 152–158, 2019.

[35] K. Kulasekarapandian, S. Jayanthi, A. Muthukumari, A. Arulsankar, and B. Sundaresan, "Preparation and characterization of PVC-PEO based polymer blend electrolytes complexed with lithium perchlorate," *Int. J. Eng. Res.*, vol. 5, no. 11, pp. 30–39, Feb. 2013.

[36] E. Andreassen, "Infrared and Raman spectroscopy of polypropylene," in *Polypropylene*, J. Karger-Kocsis, Ed. The Netherlands: Springer, 1999, pp. 320–328.

[37] V. Ludwig et al., "Analysis by Raman and infrared spectroscopy combined with theoretical studies on the identification of plasticizer in PVC films," *Vib. Spectrosc.*, vol. 98, pp. 134–138, Sep. 2018.

[38] D. Xu et al., "A gold nanoparticle doped flexible substrate for microplastics SERS detection," *Phys. Chem. Chem. Phys.*, vol. 24, no. 19, pp. 12036–12042, Apr. 2022.

[39] S. L. Wright, J. M. Levermore, and F. J. Kelly, "Raman spectral imaging for the detection of inhalable microplastics in ambient particulate matter samples," *Environ. Sci. Tech.*, vol. 53, no. 15, pp. 8947–8956, 2019.

[40] Y. Du et al., "Raman spectroscopy-based adversarial network combined with SVM for detection of foodborne pathogenic bacteria," *Talanta*, vol. 237, 2022, Art. no. 122901.

[41] S. Khan et al., "Analysis of dengue infection based on Raman spectroscopy and support vector machine (SVM)," *Biomed. Opt. Exp.*, vol. 7, no. 6, pp. 2249–2256, Jun. 2016.

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.

[42] X. Zheng et al., "Raman spectroscopy for rapid and inexpensive diagnosis of echinococcosis using the adaptive iteratively reweighted penalized least squares-Kennard–stone-back propagation neural network," *Laser Phys Lett*, vol. 15, no. 8, Jun. 2018, Art. no. 085702.
[43] H. Zhang et al., "Rapid identification of cervical adenocarcinoma, and cervical squamous cell carcinoma tissue based on Raman spectroscopy combined with multiple machine learning algorithms," *Photodiagn. Photodyn.*, vol. 33, Mar. 2021, Art. no. 102104.
[44] F. Tian, F. Tan, and H. Li, "An rapid nondestructive testing method for distinguishing rice producing areas based on Raman spectroscopy and support vector machine," *Vib. Spectrosc.*, vol. 107, Mar. 2020, Art. no. 103017.

Yinlong Luo is currently working toward the master's degree with Hohai University, Changzhou, China. His research interests include Raman spectroscopy detection, and machine learning.

Wei Su received the Doctoral degree from the Nanjing University of Science and Technology, Nanjing, China, in 2016. He is currently with Hohai University, Nanjing, as an Associate Professor. His research interests include nano-photonic materials and devices, and SERS detection in aquatic biomass.

Xiaobin Xu received the Doctoral degree from the Nanjing University of Science and Technology, Nanjing, China, in 2017. He is with Hohai University, Nanjing, as an Associate Professor. His research focuses on applications of deep learning.

Dewen Xu is currently working toward the master's degree with Hohai University, Changzhou, China. His research interests include surface enhanced Raman spectroscopy detection, and machine learning.

Zhenfeng Wang is currently working toward the master's degree with Hohai University, Nanjing, China. His research interests include Raman spectroscopy detection, and machine learning.

Hong Wu received the Doctoral degree from the Nanjing University of Science and Technology, Nanjing, China, in 2015. She is with the Nanjing University of Posts & Telecommunications, Nanjing, as a Lecturer. Her research interests include optoelectronic integration devices and optical sensing devices.

Bingyan Chen received the Doctoral degree from Hohai University, Nanjing, China, in 2016. He is currently with Hohai University, as a Professor. His research interests include Raman detection, discharge plasmas, and their applications.

Jian Wu received the Doctoral degree from Shanghai Jiao Tong University, Shanghai, China, in 2015. He is currently with the National University of Defense Technology, as an Associate Researcher. His research interests include surface enhanced Raman spectroscopy, and nano-photonic materials and devices.

Authorized licensed use limited to: University of Michigan Library. Downloaded on November 10,2026 at 19:44:57 UTC from IEEE Xplore. Restrictions apply.